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# A Mediator-Free Electroenzymatic Sensing Methodology to Mitigate Ionic and Electroactive Interferents' Effects for Reliable Wearable Metabolite and Nutrient Monitoring

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**Wearable electroenzymatic sensors enable monitoring of clinically informative biomolecules in epidermally retrievable biofluids. Conventional wearable enzymatic sensors utilize Prussian Blue (a redox mediator) to achieve selectivity against electroactive interferents. However, the use of Prussian Blue presents fundamental challenges including: 1) the susceptibility of the sensor response to dynamic concentration variation of ionic species and 2) the poor operational stability due to the degradation of its framework. As an alternative wearable electroenzymatic sensor development methodology to bypass the aforementioned limitations, a mediator-free sensing interface is devised, comprising of a coupled platinum nanoparticle/multiwall carbon nanotube layer and a permselective membrane. The interface is adapted to develop sensors targeting glucose, lactate, and choline (as examples of informative metabolites and nutrients), showing high degrees of sensitivity, selectivity (against a wide panel of naturally present and diverse interfering species), stability (<6.5% signal drift over 20 h operation), and reliability of sensing operation in sweat samples. By integration within a readout board, a wireless sample-to-answer system is realized for on-body sweat biomarker analysis. This methodology can be adapted to target a wide panel of biomarkers in various biofluids, introducing a new sensor development direction for personal health monitoring.**

## 1. Introduction

Wearable biomarker sensors present great potential for transforming personalized healthcare and precision medicine, because they allow frequent and convenient harvesting of relevant physiological data from noninvasively accessible biofluid samples such as sweat.[1–9] This information can be used to provide actionable feedback to the user with respect to adopting/modifying lifestyle routines and daily activities such as nutrition and physical exercise. To this end, tracking circulating metabolites (e.g., glucose and lactate) and nutrients (e.g., choline) plays a significant role in rendering a complete view of the body's dynamic chemistry.

To measure these molecules in epidermally retrievable biofluids, electrochemical sensing interfaces are suitable, because of their simplicity, low cost, and ease of integration with readout electronics to realize sample-to-answer sensing systems.[10,11] To this end, wearable electroenzymatic sensors (a subclass of electrochemical sensors) have been developed to measure biomarkers such as glucose and lactate in sweat.[1–4] The underlying sensing mechanism for the majority of the demonstrated sensing interfaces is based on leveraging enzymatic reactions for the simultaneous production/electroanalytical quantification of hydrogen peroxide ($H_2O_2$), which is generated in proportional amounts to the target molecule's concentration.[12] Direct amperometric detection of $H_2O_2$, using conventional electrodes (e.g., gold, Au, and platinum, Pt) is

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practically achievable at ≈0.6 V (vs silver/silver chloride, Ag/AgCl).[13] At this potential, various other electroactive species (e.g., ascorbic acid, AA, and uric acid, UA) can also be oxidized at the electrode surface together with $H_2O_2$ and interfere in the measurement.[14,15] To address this issue, previously reported wearable enzymatic sensors commonly incorporate a redox mediator-based layer (e.g., Prussian Blue, PB, also referred to as "artificial peroxidase") to lower the applied potential (close to ≈0 V) for hydrogen peroxide detection.[16]

However, as widely reported in the context of conventional electrochemical sensors, PB-based sensing presents fundamental shortcomings centering on confounding effects due to variation in biofluid conditions, rendering it unsuitable (at current stage) for the envisioned wearable applications that target biofluids with dynamically varying ionic strength and pH.[1,17–24] In particular, PB-based wearable sensor responses can be distorted by naturally occurring variation in ionic species' concentration in biofluids, because they rely on ionic species (such as potassium, $K^+$) for charge compensation and facilitating the intended PB-based redox reactions (i.e., reduction to Prussian White, PW, from PB, as conceptualized in **Figure 1**a).[17–20] Furthermore, variation in biofluid pH (especially around neutral pH) can lead to the degradation of the PB framework resulting in the loss of its electrocatalytic activity.[21–24] Thus, while electroenzymatic sensing has been realized using PB, its inherent limitations warrant focused efforts aimed at resolving them, or alternatively, necessitate exploration of other sensing methodologies that by design are suitable for sensing operation across diverse biofluid environments with varying compositions.

Here, first, the inherent limitations of PB were studied and characterized in terms of sensor selectivity against nonelectroactive ionic species and operational stability. Then, as an alternative methodology for the development of enzymatic sensors well-suited for wearable biofluid sensing, we devise a mediator-free sensing interface (Figure 1b). To achieve high sensor selectivity, sensitivity, and stability, our approach leverages: 1) a permselective membrane (poly-*m*-phenylenediamine, PPD) to reject interfering electroactive species;[25,26] 2) a Pt-based electrode, which is naturally inert against ionic species (unlike PB); and 3) a $H_2O_2$ electroanalysis layer with Pt nanoparticles (PtNP)

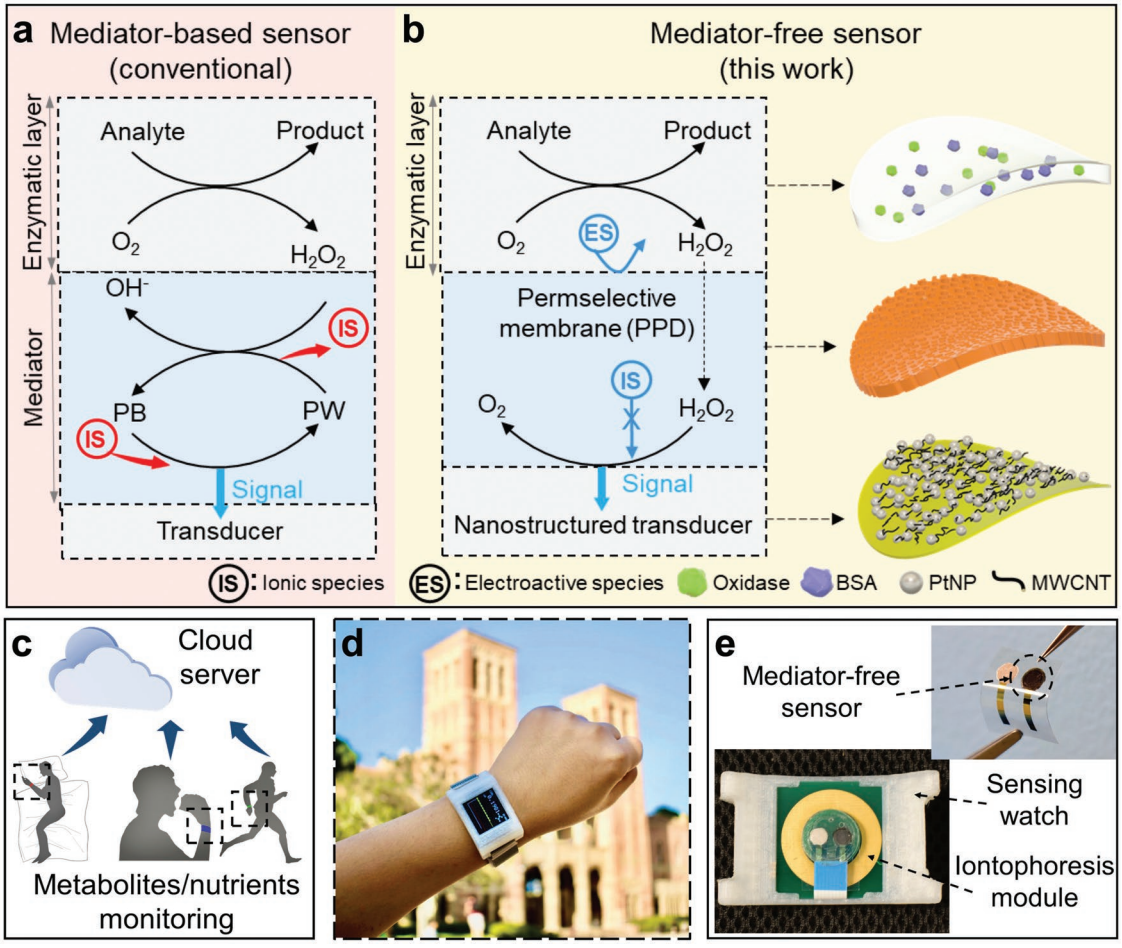



**Figure 1.** A mediator-free electroenzymatic wearable sensing system. a) Reaction schematic of a conventional mediator (PB)-based electroenzymatic sensor. b) Reaction schematic of the devised mediator-free electroenzymatic sensor. The sensor interface includes an enzymatic layer, a permselective membrane (PPD), and a MWCNT/PtNP-based $H_2O_2$ electroanalysis layer. c) Conceptual illustration of different application scenarios for the devised sensing system, which allows for wireless metabolite and nutrient data acquisition to transmit to a cloud server. d,e) A complete wearable sensing system in the form of a smartwatch, consisting of a mediator-free sensor interfacing a readout circuit board (with an optional iontophoresis module for on-demand sweat induction).


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electrodeposited onto a multiwall carbon nanotube (MWCNT) to achieve superior electrocatalytic capability toward the detection of $H_2O_2$.[27,28]

To demonstrate the versatility of the methodology to target a panel of analytes, as example critical metabolites and nutrients accessible in biofluids such as sweat, we adapted the sensing interfaces to measure glucose, lactate, and choline through immobilizing their respective oxidases. These compounds were selected in particular because tracking their circulating concentration levels can enable actionable feedback to the user and trigger active interventions within a short period, assuming timely notification. For example, sweat glucose has been shown, at least qualitatively, to track blood glucose levels.[29–31] Feedback from noninvasive wearable sensors indicating abnormally high or low levels could enable the awareness and lifestyle adjustment of general users. Similarly, sweat lactate is speculated to be informative of physiological status (e.g., physical stress and aerobic/anaerobic transitions) of individuals.[32,33] Tracking its presence in sweat during a physical activity may inform personalized and real-time performance optimization strategies. Finally, choline is important for general organ function and serves as a precursor to the neurotransmitter acetylcholine.[34,35] Short-term deficiency of choline, which can cause fat accumulation in the liver and other effects, could be reversed within a few days by reintroducing choline to the individual's diet.[34,35]

The devised mediator-free-based sensors were comprehensively characterized from the standpoints of sensitivity, selectivity, and stability, to demonstrate their suitability for the envisioned wearable applications (conceptualized in Figure 1c). As shown in Figure 1d,e, by integration within a custom-developed smartwatch, containing an optional sweat stimulating iontophoresis module, a wireless sensing system is realized for on-body sweat biomarker data acquisition and transmission to the cloud server. The versatility of the demonstrated methodology allows for its use to target a wide panel of biomarkers in various biofluids, introducing a new sensor development direction for personal health monitoring applications.

## 2. Results and Discussion

### 2.1. Characterization of PB-Based Sensors' Limitations for Sweat-Based Analysis

The core functionality of the electroenzymatic sensor is the sensitive and selective detection of $H_2O_2$, which is the byproduct of the oxidase-based reactions. However, sensitive detection of $H_2O_2$ with conventional electrodes (e.g., Pt and Au) usually requires a high applied potential (e.g., >0.6 V vs Ag/AgCl).[13] At this potential, electroactive interferents (e.g., UA and AA) in the biofluid matrix can also be oxidized and could mask the desired $H_2O_2$ signals.[14,15] To lower the applied potential, the majority of wearable electroenzymatic sensors rely on Prussian Blue ($KFe^{III}[Fe^{II}(CN)_6]$, considered as an "artificial peroxidase") as a mediating layer to reduce the applied potential to near 0 V, and thus, to minimize the influence of electroactive interferents.[1–4]

However, inherent limitations of the PB-based interface may make it unsuitable (at current stage) for epidermally retrieved biofluid analysis. To illustrate this point, here, following the previously reported protocols, PB-based $H_2O_2$ sensors were fabricated on Au-electrode patterned polyethylene terephthalate (PET) substrates (Figure S1 (Supporting Information) confirms the successful deposition of PB on Au).[1–3,36] Then, the shortcomings of the PB-based interface were characterized from the perspective of sensor selectivity against ionic species and operational stability.

#### 2.1.1. Sensor Selectivity against Ionic Species

Even though PB-based $H_2O_2$ sensing has been demonstrated with low applied potential to achieve selectivity against non-target electroactive species, the interfering effect of nonelectroactive ionic species has often been overlooked. The latter group of species participate in the charge compensation mechanisms for the conversion of PB to PW ($K_2Fe^{II}[Fe^{II}(CN)_6]$). The catalytic capabilities of PB are generally attributed to its unique zeolite crystal structure (a coordination complex with $Fe^{II}$ and $Fe^{III}$ ions alternating in face centers of a cubic lattice, where cyanide ions $CN^-$ act as ligands between the two iron states).[22] Given the PB lattice's characteristic channel radius of 0.16 μm, the relatively smaller nonelectroactive ionic species can diffuse to the interior of the structure, where they can intercalate among $Fe^{II}$ and $Fe^{III}$ ions and participate in redox conversion of PB to PW.[22] These ionic species include hydrated cations with smaller radii, such as $K^+$ (0.125 μm), $Rb^+$ (0.128 μm), $Cs^+$ (0.119 μm), and $NH_4^+$ (0.125 μm).[36–38] Among these cations, $K^+$ is of particular importance in wearable biomarker sensing applications due to its high abundance and inter/intrasubject variation in sweat.[39–41]

The involvement of $K^+$ in the PB-based mediation of the $H_2O_2$ reduction can be illustrated by the equations outlined below (**Figure 2**a)[16,22]

$$PB + e^- + K^+ \rightarrow PW \quad (1)$$

$$H_2O_2 + PW \rightarrow OH^- + PB + K^+ \quad (2)$$

The processes in Equation (1) and subsequently Equation (2) are directly influenced by $K^+$ concentration variation (with a response time larger than 20 s, as illustrated in Figure S2 in the Supporting Information). To investigate this effect, as shown in Figure 2b and Figure S3 (Supporting Information), cyclic voltammetry (CV) measurements were performed with the fabricated PB-based sensors in a buffer with a physiologically relevant ionic strength (here, similar to that of sweat). Figure 2c indicates that with the higher $K^+$ concentration level in the buffer, the larger subsidiary cathodic peak current level (a measure of the PB-to-PW conversion) was observed, which can lead to the conclusion that the $K^+$ concentration level has a direct effect on the PB/PW redox reaction rate.[17–20] This phenomenon is critical to the envisioned sweat sensing application, because the PB/PW redox reaction rate influences the reaction rate of $H_2O_2$ reduction, and subsequently the measured current. Accordingly, because the $K^+$ of the target sample is unknown (unless measured by an auxiliary sensor) and differs from that of the calibrator sample, pre/postcalibration-based techniques cannot be simply used to relate the measured current to the target analyte's concentration level.


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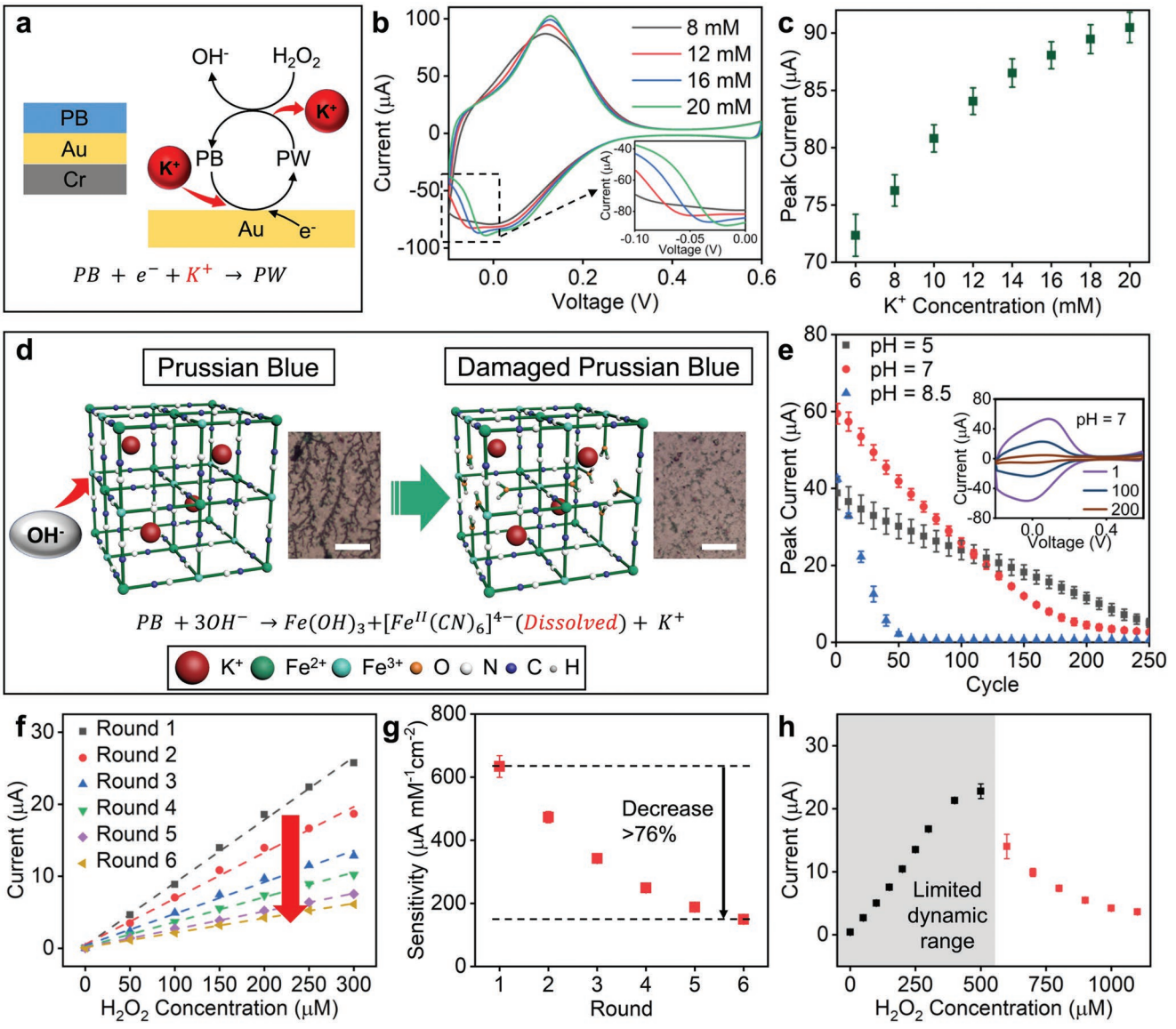



**Figure 2.** Characterization of the PB-based sensing interface's fundamental limitations. a) Reaction schematic of $H_2O_2$ sensing (right) using a PB-based sensing interface (left). b) CV diagrams generated by a representative PB-based sensing interface in response to testing buffer with different concentration levels of $K^+$ ($8 \times 10^{-3}$, $12 \times 10^{-3}$, $16 \times 10^{-3}$, and $20 \times 10^{-3}$ M, spanning across the concentration range of $K^+$ in eccrine sweat). Inset shows partial CV diagrams to highlight the increasing trend of the subsidiary cathodic peak current levels with increasing $K^+$ concentration levels in the buffer. c) CV subsidiary cathodic peak current levels versus $K^+$ concentration levels in testing buffer (generated by PB-based sensing interfaces, $n = 3$, error bars indicate standard error). d) Schematics of the crystal structures of PB and $OH^-$ damaged PB. Insets show representative optical images of PB before and after 250-cycle CV scanning (pH = 7, scale bars indicate 0.5 mm). e) CV cathodic peak current levels versus scan cycle numbers (tested in three pH conditions, 5, 7, and 8.5, generated by PB-based sensing interfaces, $n = 3$, error bars indicate standard error). Inset shows a representative PB-based sensing interface's CV diagrams at cycles 1, 100, and 200 (pH = 7). f) Six consecutive $H_2O_2$ calibration plots of a representative PB-based sensing interface. The arrow indicates the decreasing trend of current response levels. g) The illustration of decreasing $H_2O_2$ sensitivity upon calibration (six consecutive rounds of $H_2O_2$ calibrations performed with PB-based sensing interfaces, $n = 3$, error bars indicate standard error). h) Extended $H_2O_2$ calibration plot of PB-based sensing interfaces ($n = 3$, error bars indicate standard error), illustrating their limited $H_2O_2$ sensing dynamic range (shaded in gray). The testing buffer for all the experiments performed was 0.2× PBS (if unstated, with $1 \times 10^{-3}$ M $K^+$).

### 2.1.2. Operational Stability

Another fundamental limitation of PB-based sensors is its operational stability. A number of studies have reported and characterized the degradation of the PB layer in aqueous solutions, especially in alkaline environments.[21,42] Efforts have been made to improve the stability of PB structure, but the fabrication methods are typically complicated and time-consuming.[42–44] The reason for the instability of the PB structure is likely due to the strong interaction between $Fe^{III}$ ions and hydroxyl ions ($OH^-$), which can be attributed to inherent buffer conditions (especially, when operating at a pH higher than 6.4) and the $H_2O_2$ reduction.[45] To better understand this point, it is worth referring to Equation (3)[22,24]


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$$KFe^{III}\left[Fe^{II}(CN)_6\right] + 3OH^- \rightarrow Fe(OH)_3 + \left[Fe^{II}(CN)_6\right]^{4-} + K^+ \quad (3)$$

As described by Equation (3), the interaction of PB and $OH^-$ leads to the destruction of the Fe—CN—Fe bond by forming Fe–OH and the leakage of hexacyanoferrate(II) ions from the PB structure, resulting in the dissolution of the deposited PB on the electrode (Figure 2d).[46]

To characterize the operational stability of PB, continuous CV scanning (Figure 2e) was first conducted at pH = 5, 7, and 8.5 (spanning across the physiological relevant range of sweat pH).[47–49] As shown in Figure 2e, at pH = 5, the cathodic peak currents of the PB redox reaction decreased linearly over the course of the CV recordings. When changing to neutral and alkaline pH environments (pH = 7 and pH = 8.5), the peak currents decreased sharply and vanished after 150 cycles (80 min) and 50 cycles (27 min), respectively, indicating the degrading effect of the $OH^-$ interaction with PB (visualized in the inset of Figure 2d). Similar trends were also observed when immersing the fabricated PB-based sensors in the aforementioned buffers for a similar time period (where five CV measurements were performed, one every half an hour; Figure S4, Supporting Information). These results demonstrate the loss of the PB electroactivity over time, rendering PB unsuitable for $H_2O_2$-based sensing in biofluid environments.

Reduction of $H_2O_2$ also contributes to the deleterious effects of $OH^-$ on sensor stability.[16] To characterize this phenomenon, the sensitivity of $H_2O_2$ detection was evaluated over six rounds of $H_2O_2$ calibration at neutral pH. As shown in Figure 2f and Figure S5 (Supporting Information), the PB-based $H_2O_2$ sensor exhibited reduced current responses in consecutive rounds of testing. Figure 2g further illustrates that the sensor's sensitivity decreased dramatically (>76%) after six rounds of calibrations across three sensors. A distinct hysteresis behavior was also observed when measuring the PB-based sensing interface's current responses with increasing and then decreasing $H_2O_2$ concentration levels in the testing buffer (Figure S6, Supporting Information). These phenomena could be attributed to the destruction of the PB structure by $OH^-$, which, in this case, can be generated by the reduction of $H_2O_2$.

Furthermore, the dynamic range of the PB-based sensor for $H_2O_2$ detection was also evaluated. Figure 2h shows that the reduction current increased linearly with the $H_2O_2$ concentration from $50 \times 10^{-6}$ to $600 \times 10^{-6}$ M. However, a significant current drop was shown beyond $600 \times 10^{-6}$ M $H_2O_2$, which can be related to the limited availability of PB on the sensor surface as well as the degrading $OH^-$ interactions.[50] It is worth mentioning that increasing the thickness of PB layer on the sensor surface has been shown to increase the dynamic range of $H_2O_2$ detection because of the increased availability of electroactive PB on the sensor surface.[1,50] However, this increase in dynamic range comes at the expense of longer $H_2O_2$ response time and lower $H_2O_2$ detection sensitivity.[50]

### 2.2. Optimization and Characterization of PtNP-Based Sensors

Given the characterized shortcomings of PB-based sensing interfaces, as an alternative wearable enzymatic sensing approach, we devised a mediator-free electroenzymatic sensing interface, which centers on the measurement of $H_2O_2$ with high sensitivity, selectivity and stability to facilitate the reliable quantification of target biomarkers in complex biofluids.

Platinum is reported to be capable of catalyzing the oxidation of $H_2O_2$ directly with high sensitivity.[13,26] By electrochemically depositing PtNP onto the Au electrode, a $H_2O_2$ electroanalysis layer was realized to effectively oxidize $H_2O_2$ (by applying +0.5 V vs Ag/AgCl). **Figure 3**a shows the response of such layer to various $H_2O_2$ concentrations in a range from $50 \times 10^{-6}$ to $1.1 \times 10^{-3}$ M. The detectable range is twice that of the PB-based sensing interface and can be attributed to the mediator-free nature of the PtNP-based sensing interface. The PtNP-based interface exhibited minimal $H_2O_2$ detection sensitivity variations and minimal current response change through multiple rounds of calibrations, as illustrated in Figure 3b and Figure S7 in the Supporting Information (six rounds), indicating the inherent stability of the interface.

In addition, without any mediating reactions involved in the $H_2O_2$ detection, the PtNP-based sensing interface is inherently not prone to interference from variation in ionic species concentration levels in the biofluid matrix (unlike PB-based sensing). To illustrate the influence of ionic species, PB- and PtNP-based sensors were utilized to estimate the $H_2O_2$ concentration level from samples with an elevated $K^+$ level compared to that of the calibrators ($7 \times 10^{-3}$ M vs $1 \times 10^{-3}$ M). As shown in Figure 3c, the PB-based sensing interface overestimated the $H_2O_2$ concentration by 94%, whereas the PtNP-based interface enabled an accurate estimation of the $H_2O_2$ concentration. Similar superior estimation ability was obtained in glucose measurements utilizing the glucose oxidase modified sensing interface (Figure S8, Supporting Information). Given the variation of $K^+$ in biofluids (e.g., sweat), these results indicate that the PtNP-based sensing interface is expected to give more reliable readings for the envisioned wearable applications as compared to conventional PB-based sensing interfaces.

For reliable electroenzymatic analysis, the sensing interface needs to be further modified to achieve selectivity against the electroactive interfering species. To understand the context, it is worth noting that when applying +0.5 V (vs Ag/AgCl) at the working electrode, electroactive interferents in the biofluid such as UA and AA can also undergo reactions, confounding the $H_2O_2$ current response.[14,15] To this end, a permselective membrane (here, PPD) was incorporated, which only allows the passage of small and neutral molecules (e.g., $H_2O_2$) to reach the electroanalysis layer. However, the incorporated permselective membrane will inevitably act as a diffusion limiting layer as well, impairing the $H_2O_2$ detection sensitivity (Figure 3d). To regain the $H_2O_2$ detection sensitivity, a Nafion-dispersed MWCNT layer was incorporated prior to the PtNP electrodeposition, which provides a larger electrocatalytic surface area for the $H_2O_2$ electroanalysis. This new MWCNT/PtNP/PPD-based interface exhibited more than eight times the sensitivity of the PtNP/PPD-based interface (Figure 3d). A representative real-time current response of the MWCNT/PtNP/PPD-based interface to $H_2O_2$ is shown in Figure 3e. It is worth mentioning that the demonstrated sensitivity to $H_2O_2$ is superior to those reported previously.[27,51]

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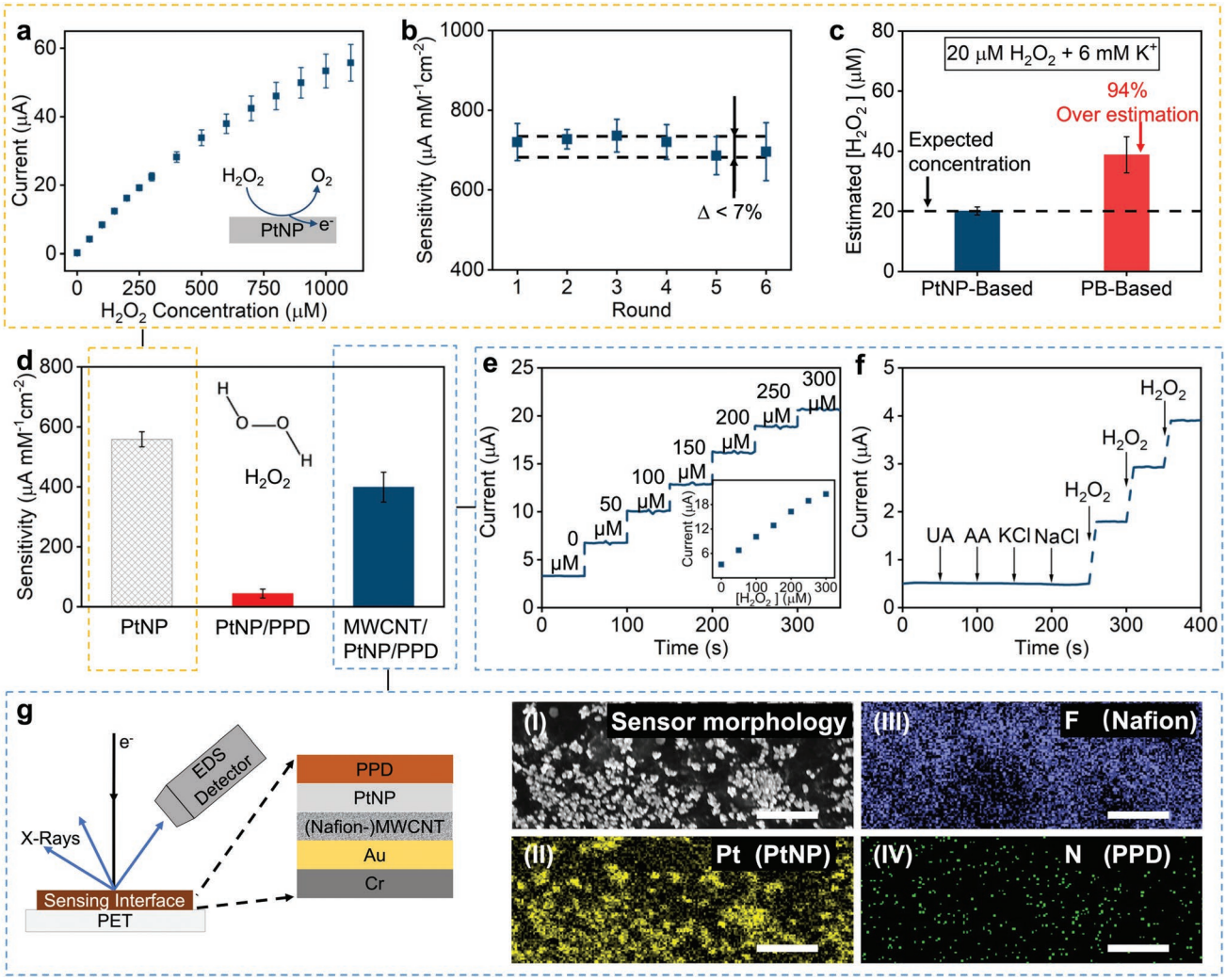



**Figure 3.** a) $H_2O_2$ calibration plot of PtNP-based sensing interfaces ($n = 3$, error bars indicate standard error, testing buffer: 0.2× PBS with $1 \times 10^{-3}$ M $K^+$). Inset shows the reaction schematic of $H_2O_2$ sensing using a PtNP-based sensing interface. b) The illustration of stable $H_2O_2$ sensitivity upon calibration (<7% change in sensitivity for six consecutive rounds of $H_2O_2$ calibrations, performed with PtNP-based sensing interfaces, $n = 3$, error bars indicate standard error, testing buffer: 0.2× PBS with $1 \times 10^{-3}$ M $K^+$). c) Estimation of the $H_2O_2$ concentration level by PB-based and PtNP-based sensing interfaces ($n = 3$ for both interfaces, error bars indicate standard error) for samples containing 20 μm $H_2O_2$ with elevated $K^+$ compared to that of the calibrators ($7 \times 10^{-3}$ M vs $1 \times 10^{-3}$ M in 0.2× PBS). d) The $H_2O_2$ sensitivities of PtNP-based (558 ± 25 μA mM$^{-1}$ cm$^{-2}$), PtNP/PPD-based (44 ± 15 μA mM$^{-1}$ cm$^{-2}$), and MWCNT/PtNP/PPD-based (399 ± 50 μA mM$^{-1}$ cm$^{-2}$) interfaces ($n = 3$ for all interfaces, error bars indicate standard error, all performed in 1× PBS). e) Real-time chronoamperometric current responses of a representative MWCNT/PtNP/PPD-based sensing interface toward $H_2O_2$ (performed in 1× PBS). Inset shows the corresponding $H_2O_2$ calibration plot. f) Selectivity study with a representative MWCNT/PtNP/PPD-based sensing interface, where its chronoamperometric current response was monitored in real-time in relation to the sequential introduction of the interferents and $H_2O_2$ (testing buffer: 1× PBS). The introduction timepoints for the interferents and $H_2O_2$ are indicated by the arrows. UA: $59 \times 10^{-6}$ M; AA: $10 \times 10^{-6}$ M; KCl: $2.4 \times 10^{-3}$ M; NaCl: $10 \times 10^{-3}$ M; $H_2O_2$: $50 \times 10^{-6}$ M. g) Left: Schematic of the EDS setup for the characterization of the (Nafion-)MWCNT/PtNP/PPD-based sensing interface. Right: I) SEM image of the developed sensing interface; EDS mappings of Pt to characterize PtNP distribution II), fluorine (F) to characterize (Nafion-)MWCNT distribution III), and nitrogen (N) to characterize PPD distribution IV). Scale bars indicate 4 μm.

To evaluate the selectivity of the MWCNT/PtNP/PPD-based interface, interfering electroactive species (UA and AA) and ions ($K^+$ and $Na^+$) were introduced (at physiologically relevant levels) to the sensor. Figure 3f shows that the recorded response to these interfering species was minimal as compared to the $H_2O_2$ response, validating the intended utility of the deposited permselective membrane and the mediator-free sensing mechanism.

To characterize the morphology of the (Nafion-)MWCNT/PtNP/PPD sensing interface and to confirm the homogenous distribution of the elements in each layer of the interface, scanning-electron microscopy coupled with energy dispersive spectroscopy (SEM-EDS) was performed (Figure 3g). As shown in Figure 3gI, the characterized particle sizes ranged from 200 to 500 nm, which falls within the expected range of the feature sizes. As further shown in Figure 3gII–IV, a


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homogeneous element distribution was observed in each functional layer.

### 2.3. Development and Characterization of Electroenzymatic Sensors Based on the MWCNT/PtNP/PPD Interface

By the oxidase-based functionalization of the MWCNT/PtNP/PPD-based interface, its excellent $H_2O_2$ sensing performance can be leveraged to develop electroenzymatic sensors targeting various metabolites and nutrients. Here, specifically, this interface is adapted to target glucose, lactate, and choline by the modification of the interface with the respective oxidases. **Figure 4**a–c shows the corresponding real-time chronoamperometric current responses of representative glucose, lactate, and choline sensors (working electrode diameter: 4 mm) within the physiological relevant concentration ranges of the respective analytes (glucose: 0–300 × $10^{-6}$ M; lactate: 0–20 × $10^{-3}$ M; and choline: 0–300 × $10^{-6}$ M).[1,52,53] Linear relationships between the sensor current responses and target analyte concentration levels were observed for all the demonstrated sensors, with measured sensitivities of 59 ± 12 μA $mM^{-1}$ $cm^{-2}$ for glucose sensors, 0.79 ± 0.18 μA $mM^{-1}$ $cm^{-2}$ for lactate sensors, and 68 ± 5 μA $mM^{-1}$ $cm^{-2}$ for choline sensors (all with high degrees of reproducibility and reversibility, as shown in Figures S9 and S10 (Supporting Information), respectively).

It is worth noting that, even though the sensing mechanisms of the demonstrated sensors are all based on the $H_2O_2$ sensing, the functionalization methods are different in terms of the use of different oxidases and diffusion limiting layers (in the case of the lactate sensor), all of which may affect the generation rate of $H_2O_2$ and the amount of $H_2O_2$ that can approach the electroanalysis layer (all sensors' schematics in Figure S11 in the Supporting Information). In the context of sweat lactate sensing, due to the fact that Michaelis constant for lactate oxidase is only ≈1 × $10^{-3}$ M, while sweat lactate concentration level is exceptionally high (≈5–20 × $10^{-3}$ M), without any provision, the sensor signal saturates.[1,52,54] Therefore, here, we incorporated a diffusion-limiting layer (i.e., polyvinyl chloride, PVC) which allows for the extension of the lactate sensor dynamic range to span across the physiological concentration range of sweat lactate.[4,54]

In order to ensure the sensors' reliable performance in complex biofluids, comprehensive selectivity tests were conducted by measuring chronoamperometric current responses of the developed sensors toward a diverse panel of interfering species within their physiological concentration ranges in sweat. As listed in Figure 4d, this panel of interfering species includes small molecules, ionic species, electroactive species, a sweat agonist, xenobiotic drugs, and a protein. As shown in Figure 4e–g, the developed glucose, lactate, and choline sensors showed negligible current responses toward all the interfering species, and significant stepwise current increases were recorded upon the introduction of the sensor-specific target molecules.

Long-term stability of the electroenzymatic sensor was also examined, and accordingly, the response of an exemplary glucose sensor was recorded over a 20 h period, while different glucose concentration levels were introduced. As shown in Figure 4h, throughout the entire test period, the sensor maintained a relatively constant response for each glucose concentration level with minimal current drift (<6.5%), indicating the excellent long-term stability and robustness of the developed sensor.

### 2.4. Ex Situ and In Situ Sweat Analysis Using Mediator-Free Electroenzymatic Sensing

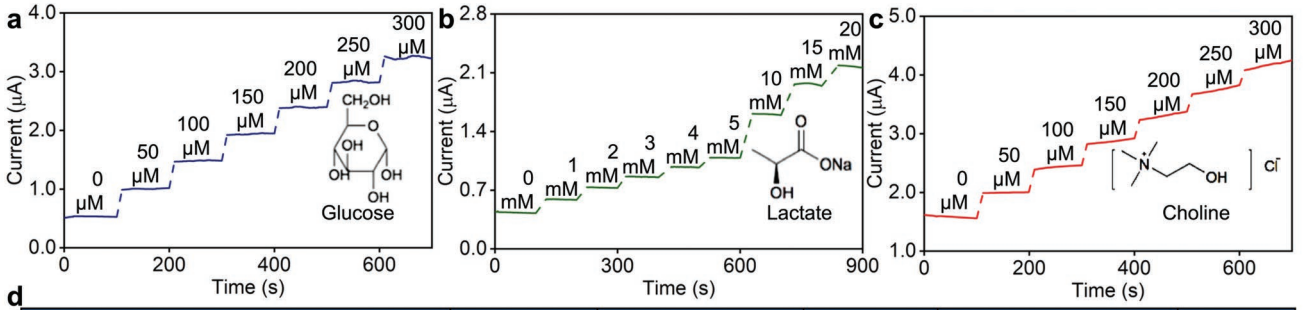
The utility of the mediator-free electroenzymatic sensing for glucose, lactate, and choline quantification was validated by performing electrochemical analysis in artificial sweat samples (five per sensor) and untreated human sweat samples (iontophoretically or exercise-induced, three per sensor), and comparing the results with expected values. As shown in **Figure 5**a–c, the sensor-estimated glucose, lactate, and choline concentration levels closely matched the expected values ($R^2_{Glucose}$ = 0.99, $R^2_{Lactate}$ = 0.80, and $R^2_{Choline}$ = 0.98). The results collectively indicate the suitability of the devised mediator-free sensing approach to quantify metabolites and nutrients in complex biofluids.

To demonstrate the suitability of the developed sensors for wearable applications, an electrochemical enzymatic sensing system was developed in the form of a smartwatch (Figure 5d; Figure S12, Supporting Information), which consists of 1) an optional iontophoresis module which can apply current on-demand to deliver sweat agonist (pilocarpine) underneath the skin for sweat stimulation (Figure S12, Supporting Information); 2) a microfluidic interface for sweat sampling and routing to the sensor surface (Figure S13, Supporting Information); 3) a mediator-free electroenzymatic sensor for biomarker analysis (glucose and lactate as example target biomarkers); 4) a custom-developed wireless readout circuitry for signal acquisition, processing, and wireless data communication to a mobile application (Figure S12, Supporting Information); and 5) an liquid-crystal display (LCD) for real-time data display.

The smartwatch was first coupled with the glucose sensor and utilized to monitor the sweat glucose change in three subjects before and after consumption of mixed meals (where standard iontophoresis protocol was performed 2 h before and 0.5 h after meal consumption in order to induce and subsequently analyze sweat). Figure 5e–g demonstrates the elevation of sweat glucose concentration level after meal consumption (in all three subjects), which is in agreement with a trend observed previously.[3,55,56]

Additionally, the smartwatch was coupled with the lactate sensor and used for continuous sweat lactate profile monitoring of a subject engaged in constant-load stationary cycling. As illustrated in Figure 5h (solid green), a decreasing profile of sweat lactate current responses were recorded, similar to that observed previously.[4,57] The control experiment conducted on the same subjects with a nonfunctionalized system (without lactate-oxidase modification) yielded the recording of minimal current response (Figure 5h, dashed black). Altogether, the on-body studies illustrate the suitability of the mediator-free electroenzymatic sensing for wearable applications.


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d

| Small molecules | | | | Ionic species | | Electroactive species | | Sweat agonist | Drugs | | Protein |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Glucose | 2. Lactate | 3. Creatinine | 4. Choline | 5. KCl | 6. NaCl | 7. UA | 8. AA | 9. Pilocarpine | 10. Aspirin | 11. Metformin | 12. Albumin |
| 50 μM | 5 mM | 84 μM | 1.6 μM | 2.4 mM | 10 mM | 59 μM | 10 μM | 50 μM | 100 μM | 774 μM | 1 nM |

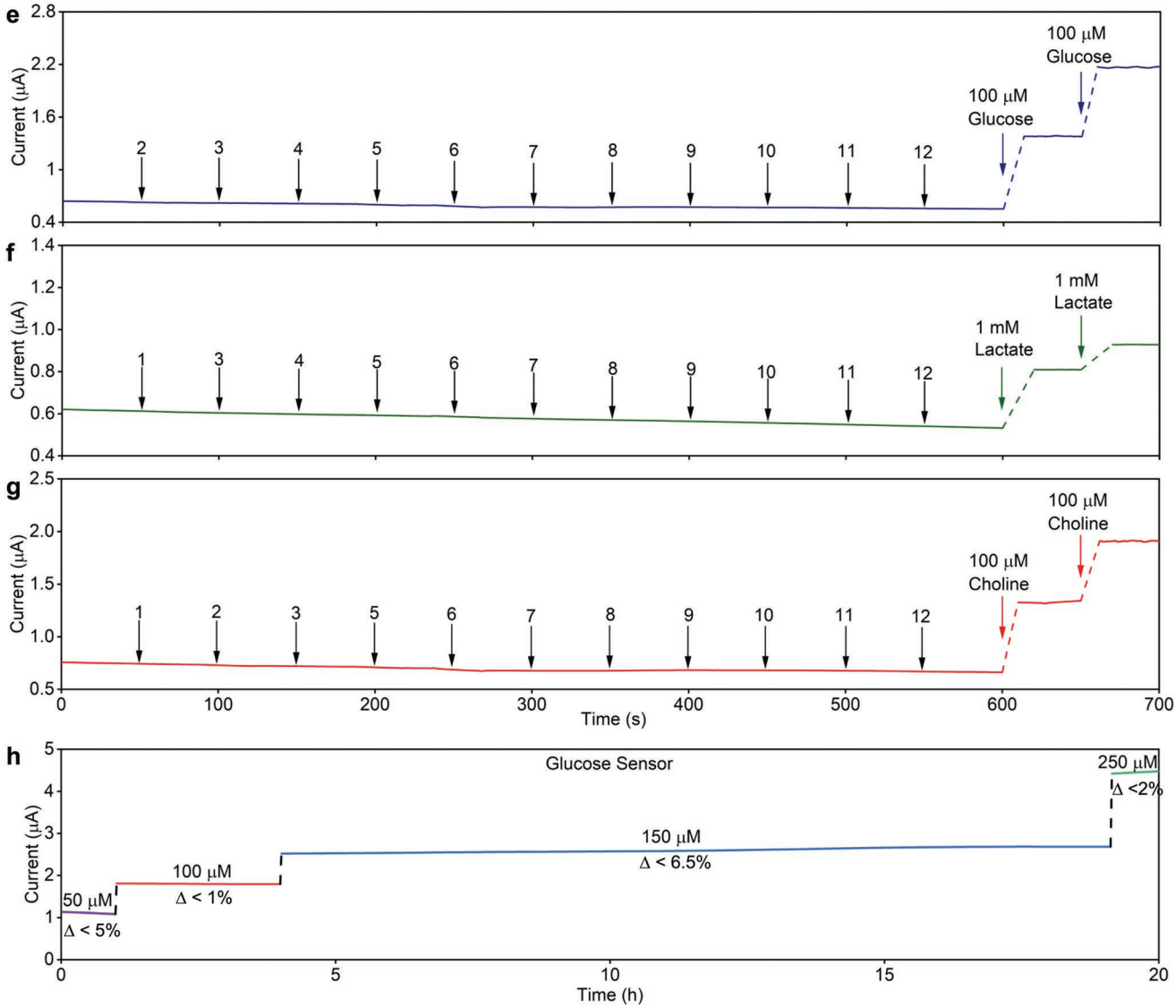



**Figure 4.** Real-time chronoamperometric current responses to target analytes for a glucose sensor a), lactate sensor b), and choline sensor c). d) Table of common interferents in biofluids (e.g., sweat). Comprehensive selectivity studies for glucose e), lactate f), and choline g) sensors, performed by chronoamperometric measurements to monitor the corresponding sensors' responses to the sequential introduction of the listed interferents and target analytes (the introduction timepoints for the interferents and target analytes are indicated by the arrows). h) A 20 h long chronoamperometric measurement performed with a representative glucose sensor, indicating the sensor's excellent long-term operational stability (the glucose concentration and the sensor current drift are annotated for the corresponding time windows). The testing buffer for all the experiments performed was 1× PBS.


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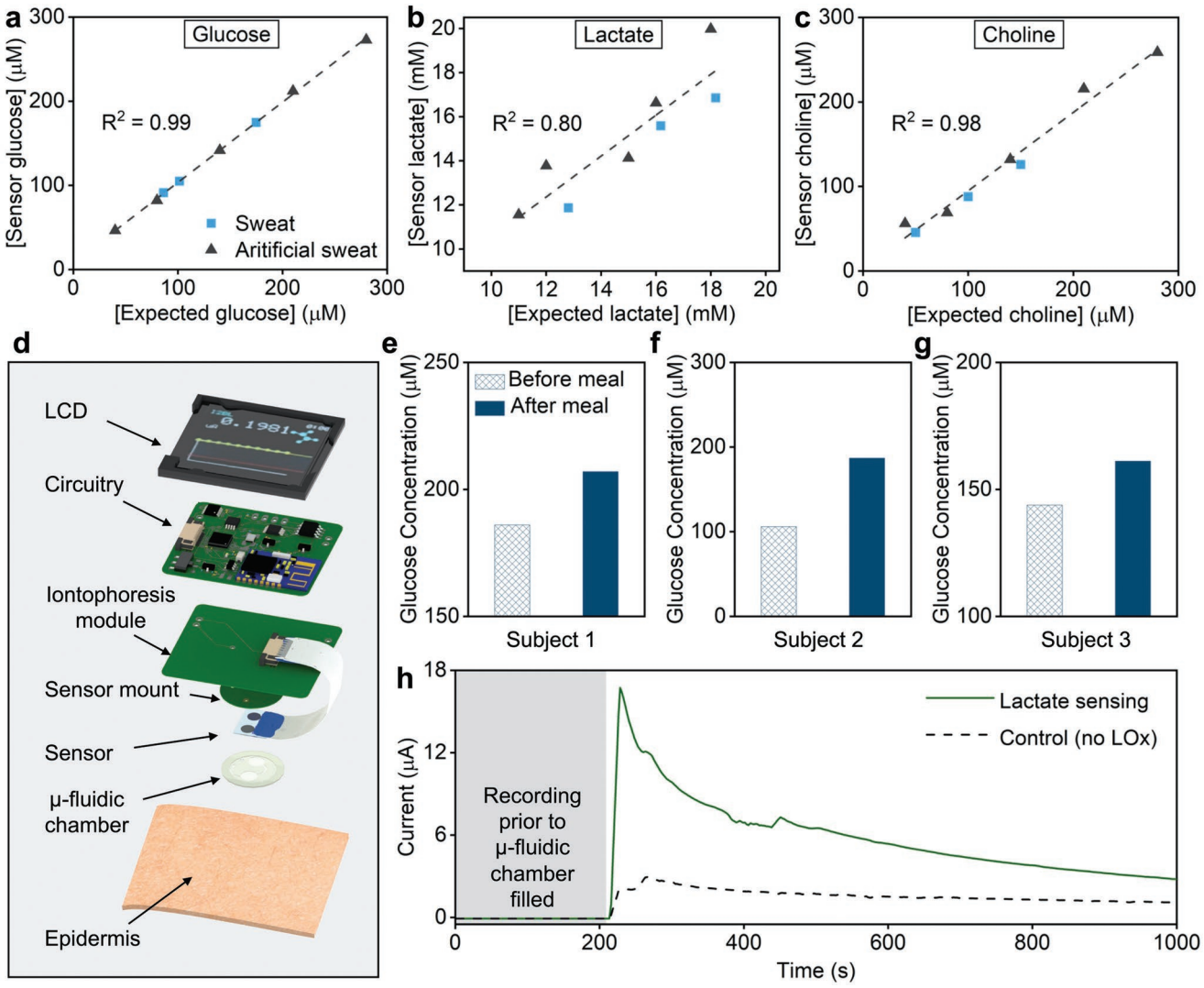



**Figure 5.** The ex situ sensor-measured glucose a), lactate b), and choline c) concentration levels in artificial sweat and untreated human sweat samples versus the corresponding expected concentration levels of the analytes. The analytes in untreated human sweat samples were quantified by a lab instrument (YSI analyzer for glucose and lactate) or spiked by known amounts (for choline). d) Exploded view of the mediator-free sensing system, indicating the coupling of the sensor with a custom-developed smartwatch. e–h) On-body monitoring of subjects' metabolites concentration levels and dynamic changes through various daily events (meal consumption and physical exercise) in different settings (sedentary and active). Measured glucose concentration levels in iontophoretically induced sweat samples of three subjects before and after consumption of mixed meals e–g). Continuous sweat lactate measurement of a subject engaged in stationary cycling h). The solid green line shows the measurement with the lactate oxidase-functionalized sensing system and the control experiment is illustrated with dashed black line, which was performed with a nonfunctionalized system (without lactate oxidase functionalization).

## 3. Conclusion

In this study, we characterized the performance of PB-based sensing to demonstrate its limitations in terms of: 1) sensor response susceptibility to ionic species (e.g., $K^+$), the concentration levels of which dynamically vary in biofluids (e.g., such as sweat), practically leading to severe inaccurate estimation of the end-product of the enzymatic reaction (i.e., $H_2O_2$), and subsequently, the target biomarkers concentration levels, and 2) degradation of the PB framework (i.e., zeolite crystal structure) caused by $OH^-$, resulting in poor operational stability. These results indicate that either significant efforts are required to further study and optimize PB-based sensors for reliable wearable biofluid analysis or alternative sensor development methodologies may need to be adopted. Aligned with the latter point, a mediator-free electroenzymatic sensing methodology is devised, which centers on utilizing a sensing interface comprised of a coupled MWCNT/PtNP layer and a permselective membrane (PPD). Through functionalization with appropriate oxidases, this interface was used to develop glucose, lactate, and choline sensors. The developed sensors were comprehensively characterized from the standpoints of sensitivity, selectivity, stability, and reliability of sensing operation in untreated sweat samples. To further demonstrate their suitability for the


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envisioned wearable applications, the glucose and lactate sensors were integrated with a custom-developed smartwatch, rendering on-body sweat biomarker data acquisition.

By tuning the materials of the devised functional layers (e.g., permselective membrane or the enzyme layer) as well as by utilizing auxiliary (e.g., temperature and pH) sensors, the demonstrated mediator-free sensing methodology can be adapted to accurately measure a broad range of biomarkers in various biofluids. In this way, the presented methodology can serve as a basis for the future wearable electroenzymatic sensor development efforts toward personal health monitoring applications.

## 4. Experimental Section

*Materials and Reagents*: Multiwalled carbon nanotubes (MWCNT, carboxylic acid functionalized), Nafion perfluorinated resin solution (5 wt%), chloroplatinic acid hexahydrate ($H_2PtCl_6 \cdot 6H_2O$), formic acid, *m*-phenylenediamine, bovine serum albumin (BSA), glutaraldehyde solution (25 wt%), polyvinyl chloride, tetrahydrofuran (THF), D-(+)-glucose, sodium L-lactate, choline chloride, acetylsalicylic acid (aspirin), creatinine, L-ascorbic acid, potassium hexacyanoferrate(II) trihydrate, potassium hexacyanoferrate(III), iron(III) chloride, potassium chloride, sodium chloride, uric acid, pilocarpine hydrochloride, metformin hydrochloride, albumin from human serum, sodium hydroxide, glucose oxidase, and choline oxidase were purchased from Sigma-Aldrich (MO, USA). Water (Optima LC/MS Grade), *2*-propanol, hydrochloric acid, Honeywell hydrogen peroxide solution (30 wt%), and phosphate buffered saline (1×, Gibco PBS, pH 7.2) were purchased from Fisher Scientific (MA, USA). Lactate oxidase was purchased from Toyobo USA, Inc. (NY, USA). Ag/AgCl ink was purchased from Ercon Incorporated (MA, USA). PILOGEL discs were purchased from ELITechGroup Inc. (UT, USA). PET (100 µm thick) was purchased from MG Chemicals (BC, Canada). Double-sided tape (170 µm thick, 9474LE 300 LSE) and Tegaderm Film were purchased from 3M Science (MN, USA).

*Biosensor Fabrication*: The biosensors were fabricated on a pair of Au electrodes (diameter: 4 mm; 30 nm Cr/100 nm Au) deposited and patterned on a PET substrate. The reference electrode was fabricated by depositing 3.2 µL of Ag/AgCl ink on the Au electrode, then dried on a hot plate at 70 °C for 20 min.

To fabricate PB-based sensors, PB was electrochemically deposited onto the Au electrode by CV from 0 to 0.5 V (vs Ag/AgCl) for 1 cycle at 20 mV $s^{-1}$ in a fresh PB solution containing $2.5 \times 10^{-3}$ M $FeCl_3$, $2.5 \times 10^{-3}$ M $K_3[Fe(CN)_6]$, $100 \times 10^{-3}$ M KCl, and $100 \times 10^{-3}$ M HCl. After the deposition, the sensors were dried on a hot plate at 100 °C for 1 h.

To fabricate PtNP-based sensors, PtNP were electrochemically deposited onto the Au electrode by chronoamperometry at –0.1 V (vs Ag/AgCl) for 10 min in a fresh Pt solution (containing $2.5 \times 10^{-3}$ M $H_2PtCl_6$ and $1.5 \times 10^{-3}$ M formic acid).[58,59] A PPD layer was subsequently electrochemically deposited onto the Au/PtNP electrode by applying 0.85 V (vs Ag/AgCl) for 120 s in a fresh PBS solution (1×) containing $5 \times 10^{-3}$ M *m*-phenylenediamine.

To fabricate the MWCNT/PtNP-based sensors, a multiwalled carbon nanotube solution was first prepared by mixing MWCNT (2 mg $mL^{-1}$) with 5 wt% Nafion solution followed by ultrasonic agitation for over 30 min to form a viscous solution of carbon nanotubes. 3.2 µL of the MWCNT solution was drop-cast onto the Au electrode and dried in the ambient environment. PtNP were later electrochemically deposited onto the MWCNT deposited Au electrode via CV from –1.0 to 0 V (vs Ag/AgCl) for 20 cycles at 50 mV $s^{-1}$ in the fresh Pt solution. A PPD layer was electrochemically deposited onto Au/MWCNT/PtNP following the same method. In order to prepare the biosensors, the electrode was further functionalized with an enzymatic layer. The BSA stabilizer solution was prepared by adding 0.8% (v/v) of 25 wt% glutaraldehyde solution into a fresh PBS solution containing 10 mg $mL^{-1}$ BSA. To develop the glucose sensor, the aforementioned BSA stabilizer solution was mixed thoroughly with a glucose oxidase solution (50 mg $mL^{-1}$ in PBS, pH 7.2) at a ratio of 1:1 (v/v). By drop casting 3.2 µL of the glucose enzyme mixture onto the Au/MWCNT/PtNP/PPD electrode, the glucose sensing interface was realized. To realize the choline sensor, the BSA stabilizer solution was mixed thoroughly with a choline oxidase solution (10 mg $mL^{-1}$ in PBS, pH 7.2) at a ratio of 1:1 (v/v). The choline sensing interface was constructed by drop-casting 3.2 µL of the choline enzyme mixture onto the Au/MWCNT/PtNP/PPD electrode. To create the lactate sensor, the BSA stabilizer solution was mixed thoroughly with a lactate oxidase solution (50 mg $mL^{-1}$ in PBS, pH 7.2) at a ratio of 1:1 (v/v). 3.2 µL of the mixture was deposited onto the Au/MWCNT/PtNP/PPD electrode and dried at room temperature, followed by dip-coating of PVC solution (3 wt% in THF) to form a lactate diffusion limiting layer. Sensors were allowed to dry overnight at 4 °C, while being protected from light. The sensors were stored at 4 °C when not in use.

*Microfluidic Interface Construction*: The microfluidic interface was created by laser-cutting (Epilog Mini 24, Epilog Laser) 2D patterns within the double-sided tape and Tegaderm Film. The inlet was created by laser-cutting holes on the PET layer. The outlet was created by laser-cutting holes on the double-sided tape backing layer. The microfluidic interface was then assembled by attaching one layer of prepatterned double-sided tape with a prepatterned transparent PET film sheet, and one layer of prepatterned Tegaderm Film with proper alignment.

*Biosensor Characterization and Analytes Measurement*: To characterize the developed PB-based sensing interface, CV measurements were conducted in 0.1 M KCl solution from –0.2 to +0.4 V at a scan rate of 50 mV $s^{-1}$. Constant potential amperometric measurements were conducted in constantly stirred PBS buffer (1× or 0.2× diluted with $1 \times 10^{-3}$ M KCl) at 0 V (vs Ag/AgCl). By stepwise addition of different concentrations of $H_2O_2$ in the PBS buffer, a series of calibration plots were obtained. To characterize the $K^+$ responses of the PB-based sensing interface, CV measurements were conducted in 0.2× diluted PBS from –0.1 to +0.6 V (vs Ag/AgCl) at a scan rate of 50 mV $s^{-1}$. Before each measurement, different concentrations of KCl were introduced into the PBS buffer.

To characterize the developed PtNP-based sensing interface, chronoamperometric measurements were conducted in constantly stirred PBS buffer (1× PBS or 0.2× diluted PBS with $1 \times 10^{-3}$ M KCl) at +0.5 V (vs Ag/AgCl). By stepwise addition of different concentrations of the targets (e.g., $H_2O_2$, glucose, lactate, or choline) in PBS buffer with corresponding sensors, a series of calibration plots were obtained. The limit of detection (LOD) for each amperometric sensing interface was calculated as LOD = 3*SD/(slope of calibration curve), where SD is the standard deviation of the baseline noise in the blank solution and slope is defined as amperometric current change versus target concentration change. Sensitivity is calculated as the slope divided by sensor's surface area. The sensor selectivity tests were conducted by stepwise addition of different interferents into the PBS, including glucose/lactate/choline, creatine, KCl, NaCl, UA, AA, pilocarpine, aspirin, metformin, and albumin. The target analytes were also introduced into the solutions after the interferents.

*Glucose and Lactate Quantification with Lab Instrument*: The glucose and lactate concentration levels from untreated human sweat samples were directly quantified with a YSI 2900 analyzer (YSI Incorporated, OH, USA).

*Characterization of Sensors Morphologies and Chemical Composition Distributions*: The sensor morphologies were characterized by field emission scanning electron microscopy (FE-SEM) (ZEISS Supra 40VP, Germany). Chemical compositions distributions were characterized by energy dispersive spectroscopy (EDS) (Thermo NORAN System 6, USA).

*On-Body Tests Using the Mediator-Free Sensor-Integrated Smartwatch*: The mediator-free sensor was transferred onto the disk contact located at the back of the custom-developed sensing smartwatch. Then, the microfluidic interface was transferred on top of the sensor before being attached to a healthy human subject's wrist. The smartwatch case was customized and 3D printed to consolidate the smartwatch components, which include LCD display, battery, printed circuit board (PCB) module, sensor, and microfluidic interface. To induce sweat by iontophoresis, a


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subject's stimulated skin area was first cleaned with deionized water and ethanol, followed by sweat gland stimulation by iontophoresis for 5 min.

*Wireless PCB Module*: The wearable PCB module is designed to activate and control the sweat stimulation and sensing circuits, and to process and communicate the obtained data wirelessly upon user commands. This circuit consists of several layers to enclose the iontophoresis and sensing modules. The main PCB section is connected to the iontophoresis module through metallic connections. This layer also encapsulates a flat flexible cable connection to interface with the sensor. Utilizing an anisotropic conductive film, the flexible cable was connected to the fabricated flexible electrodes.

The required sweat stimulation circuitry driving the iontophoresis electrodes consists of a current source, where the current applied is independent of the user's skin type (impedance). The current levels are controlled through user command and via the digital-to-analog (DAC) port of the microcontroller. This current is limited through the current limiting circuits implemented on board and constantly monitored by the microcontroller (STM8L-UFQFN20, STMicroelectronics) to apply the minimal current set by the user in the comfortable range of 0.1–1.5 mA.

After the iontophoresis process is done, the sensing mode is activated. At this stage, the microcontroller will activate the LMP91000 potentiostat chip (Texas Instruments) through I2C-controlled signals. This chip is programmed to apply +0.5 V across the working and reference electrodes connected to the sensor. Then, to filter out high-frequency interference and the user's motion artifacts, the output voltage of the potentiostat's internal transimpedance amplifier is passed through a fifth-order low-pass filter (LPF). The LPF was applied using a MAX7422 chip (Maxim Integrated) with a cut-off frequency of 1 Hz. Through the microcontroller's built-in 12 bit analog-to-digital (ADC) unit then converts the processed analog output of the LPF into the digital domain for real time biomarker monitoring which is then plotted on the LCD (st7735-TFT-LCD, Sitronix Technology Corporation) and transfers this data to the user's smartphone.

To power the PCB, a single miniaturized rechargeable lithium-ion polymer battery with a nominal voltage of 3.7 V was used. Additionally, in this design, we obtained low/ultralow power components to make the device wearable and feasible for on-body testing. For our specific application and duration of tests, we characterized that peak supply current levels on the order of 100 mA must be drawn from the battery. This requirement can vary based on different modes and durations of applications.

*Smartphone Application and Smartwatch Design*: User commands and data are bilaterally communicated between the PCB module and the custom-developed Android application through Bluetooth communication. In this application, we designed graphical interfaces for each mode to plot the output current (applied in the iontophoresis mode) and the sensing results. Additionally, the data received is time stamped, analyzed, and stored locally for future reference. This stored data can then be sent to a custom-developed Google Cloud platform for further analysis.

*Institutional Review Board (IRB) Approval for Human Subject Testing*: The conducted human subject experiments were performed in compliance with the protocols that are approved by the IRB at the University of California, Los Angeles (IRB#17-000170). All subjects gave written informed consent before participation in the study.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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## Conflict of Interest

The authors declare no conflict of interest.

## Keywords

electrochemical sensors, mediator-free sensors, metabolite monitoring, nutrient monitoring, wearable biomarker sensors

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[1] W. Gao, S. Emaminejad, H. Y. Y. Nyein, S. Challa, K. V. Chen, A. Peck, H. M. Fahad, H. Ota, H. Shiraki, D. Kiriya, D. H. Lien, G. A. Brooks, R. W. Davis, A. Javey, *Nature* **2016**, *529*, 509.
[2] H. S. Lin, Y. C. Zhao, S. Y. Lin, B. Wang, C. Yeung, X. B. Cheng, Z. Q. Wang, T. Y. Cai, W. Z. Yu, K. King, J. W. Tan, K. Salahi, H. Hojaiji, S. Emaminejad, *Lab Chip* **2019**, *19*, 2844.
[3] S. Emaminejad, W. Gao, E. Wu, Z. A. Davies, H. Y. Y. Nyein, S. Challa, S. P. Ryan, H. M. Fahad, K. Chen, Z. Shahpar, S. Talebi, C. Milla, A. Javey, R. W. Davis, *Proc. Natl. Acad. Sci. USA* **2017**, *114*, 4625.
[4] S. Imani, A. J. Bandodkar, A. M. V. Mohan, R. Kumar, S. F. Yu, J. Wang, P. P. Mercier, *Nat. Commun.* **2016**, *7*, 11650.
[5] A. Koh, D. Kang, Y. Xue, S. Lee, R. M. Pielak, J. Kim, T. Hwang, S. Min, A. Banks, P. Bastien, M. C. Manco, L. Wang, K. R. Ammann, K. I. Jang, P. Won, S. Han, R. Ghaffari, U. Paik, M. J. Slepian, G. Balooch, Y. G. Huang, J. A. Rogers, *Sci. Transl. Med.* **2016**, *8*, 366ra165.
[6] J. Kim, A. S. Campbell, J. Wang, *Talanta* **2018**, *177*, 163.
[7] H. Y. Y. Nyein, M. Bariya, L. Kivimaki, S. Uusitalo, T. S. Liaw, E. Jansson, C. H. Ahn, J. A. Hangasky, J. Q. Zhao, Y. J. Lin, T. Happonen, M. H. Chao, C. Liedert, Y. B. Zhao, L. C. Tai, J. Hiltunen, A. Javey, *Sci. Adv.* **2019**, *5*, eaaw9906.
[8] A. Hauke, P. Simmers, Y. R. Ojha, B. D. Cameron, R. Ballweg, T. Zhang, N. Twine, M. Brothers, E. Gomez, J. Heikenfeld, *Lab Chip* **2018**, *18*, 3750.
[9] J. Kim, I. Jeerapan, S. Imani, T. N. Cho, A. Bandodkar, S. Cinti, P. P. Mercier, J. Wang, *ACS Sens.* **2016**, *1*, 1011.
[10] W. Gao, H. Y. Y. Nyein, Z. Shahpar, H. M. Fahad, K. Chen, S. Emaminejad, Y. J. Gao, L. C. Tai, H. Ota, E. Wu, J. Bullock, Y. P. Zeng, D. H. Lien, A. Javey, *ACS Sens.* **2016**, *1*, 866.

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[11] L. C. Tai, W. Gao, M. H. Chao, M. Bariya, Q. P. Ngo, Z. Shahpar, H. Y. Y. Nyein, H. Park, J. Sun, Y. Jung, E. Wu, H. M. Fahad, D. H. Lien, H. Ota, G. Cho, A. Javey, *Adv. Mater.* **2018**, *30*, 1707442.
[12] S. B. Bankar, M. V. Bule, R. S. Singhal, L. Ananthanarayan, *Biotechnol. Adv.* **2009**, *27*, 489.
[13] G. G. Guilbault, G. J. Lubrano, *Anal. Chim. Acta* **1973**, *64*, 439.
[14] A. A. Karyakin, E. E. Karyakina, L. Gorton, *Talanta* **1996**, *43*, 1597.
[15] H. Zhang, Y. J. Yin, P. Wu, C. X. Cai, *Biosens. Bioelectron.* **2012**, *31*, 244.
[16] A. A. Karyakin, E. E. Karyakina, L. Gorton, *Electrochem. Commun.* **1999**, *1*, 78.
[17] J. Q. Ang, T. T. N. Binh, C. S. Toh, *Sens. Actuators, B* **2011**, *157*, 417.
[18] V. Krishnan, A. L. Xidis, V. D. Neff, *Anal. Chim. Acta* **1990**, *239*, 7.
[19] J. Q. Ang, B. T. T. Nguyen, Y. Huang, C. S. Toh, *Electrochim. Acta* **2010**, *55*, 7903.
[20] B. T. T. Nguyen, J. Q. Ang, C. S. Toh, *Electrochem. Commun.* **2009**, *11*, 1861.
[21] F. Ricci, A. Amine, G. Palleschi, D. Moscone, *Biosens. Bioelectron.* **2003**, *18*, 165.
[22] R. Koncki, *Crit. Rev. Anal. Chem.* **2002**, *32*, 79.
[23] D. E. Stilwell, K. H. Park, M. H. Miles, *J. Appl. Electrochem.* **1992**, *22*, 325.
[24] J. M. Noel, J. Medard, C. Combellas, F. Kanoufi, *ChemElectroChem* **2016**, *3*, 1178.
[25] X. M. Wen, B. Wang, S. Huang, T. Y. Liu, M. S. Lee, P. S. Chung, Y. T. Chow, I. W. Huang, H. G. Monbouquette, N. T. Maidment, P. Y. Chiou, *Biosens. Bioelectron.* **2019**, *131*, 37.
[26] B. Wang, B. Koo, L. W. Huang, H. G. Monbouquette, *Analyst* **2018**, *143*, 5008.
[27] S. Hrapovic, Y. L. Liu, K. B. Male, J. H. T. Luong, *Anal. Chem.* **2004**, *76*, 1083.
[28] S. K. Vashist, D. Zheng, K. Al-Rubeaan, J. H. T. Luong, F. S. Sheu, *Biotechnol. Adv.* **2011**, *29*, 169.
[29] J. Moyer, D. Wilson, I. Finkelshtein, B. Wong, R. Potts, *Diabetes Technol. Ther.* **2012**, *14*, 398.
[30] Y. J. Hong, H. Lee, J. Kim, M. Lee, H. J. Choi, T. Hyeon, D. H. Kim, *Adv. Funct. Mater.* **2018**, *28*, 1805754.
[31] H. Lee, C. Song, Y. S. Hong, M. S. Kim, H. R. Cho, T. Kang, K. Shin, S. H. Choi, T. Hyeon, D. H. Kim, *Sci. Adv.* **2017**, *3*, e1601314.
[32] Z. Sonner, E. Wilder, J. Heikenfeld, G. Kasting, F. Beyette, D. Swaile, F. Sherman, J. Joyce, J. Hagen, N. Kelley-Loughnane, R. Naik, *Biomicrofluidics* **2015**, *9*, 031301.
[33] P. J. Derbyshire, H. Barr, F. Davis, S. P. J. Higson, *J. Physiol. Sci.* **2012**, *62*, 429.
[34] S. H. Zeisel, K. A. da Costa, *Nutr. Rev.* **2009**, *67*, 615.
[35] S. H. Zeisel, *J. Am. Coll. Nutr.* **1992**, *11*, 473.
[36] K. Itaya, I. Uchida, V. D. Neff, *Acc. Chem. Res.* **1986**, *19*, 162.
[37] K. Itaya, T. Ataka, S. Toshima, *J. Am. Chem. Soc.* **1982**, *104*, 4767.
[38] F. Ricci, G. Palleschi, *Biosens. Bioelectron.* **2005**, *21*, 389.
[39] A. J. Bandodkar, J. Choi, S. P. Lee, W. J. Jeang, P. Agyare, P. Gutruf, S. Q. Wang, R. A. Sponenburg, J. T. Reeder, S. Schon, T. R. Ray, S. L. Chen, S. Mehta, S. Ruiz, J. A. Rogers, *Adv. Mater.* **2019**, *31*, 1902109.
[40] M. J. Patterson, S. D. R. Galloway, M. A. Nimmo, *Exp. Physiol.* **2000**, *85*, 869.
[41] S. J. Montain, S. N. Cheuvront, H. C. Lukaski, *Int. J. Sport Nutr. Exercise Metab.* **2007**, *17*, 574.
[42] Y. J. Lin, M. Bariya, H. Y. Y. Nyein, L. Kivimaki, S. Uusitalo, E. Jonsson, W. B. Ji, Z. Yuan, T. Happonen, C. Liedert, J. Hiltunen, Z. Y. Fan, A. Javey, *Adv. Funct. Mater.* **2019**, *29*, 1902521.
[43] N. A. Sitnikova, A. V. Borisova, M. A. Komkova, A. A. Karyakin, *Anal. Chem.* **2011**, *83*, 2359.
[44] N. A. Sitnikova, M. A. Komkova, I. V. Khomyakova, E. E. Karyakina, A. A. Karyakin, *Anal. Chem.* **2014**, *86*, 4131.
[45] B. J. Feldman, R. W. Murray, *Inorg. Chem.* **1987**, *26*, 1702.
[46] A. Malinauskas, R. Araminaite, G. Mickeviciute, R. Garjonyte, *Mater. Sci. Eng., C* **2004**, *24*, 513.
[47] H. M. Emrich, E. Stoll, B. Friolet, J. P. Colombo, R. Richterich, E. Rossi, *Pediatr. Res.* **1968**, *2*, 464.
[48] F. Herrmann, L. Mandol, *J. Invest. Dermatol.* **1955**, *24*, 225.
[49] H. Y. Y. Nyein, W. Gao, Z. Shahpar, S. Emaminejad, S. Challa, K. Chen, H. M. Fahad, L. C. Tai, H. Ota, R. W. Davis, A. Javey, *ACS Nano* **2016**, *10*, 7216.
[50] J. H. Zhu, Z. Q. Zhu, Z. S. Lai, R. Wang, X. M. Guo, X. Q. Wu, G. X. Zhang, Z. R. Zhang, Z. R. Zhang, Y. T. Wang, Z. Y. Chen, *Sensors* **2002**, *2*, 127.
[51] T. Unmussig, A. Weltin, S. Urban, P. Daubinger, G. A. Urban, J. Kieninger, *J. Electroanal. Chem.* **2018**, *816*, 215.
[52] M. Bariya, H. Y. Y. Nyein, A. Javey, *Nat. Electron.* **2018**, *1*, 160.
[53] S. S. Craig, S. A. S. Craig, M. S. Ganio, C. M. Maresh, H. Greg, K. A. da Costa, S. H. Zeisel, *J. Int. Soc. Sports Nutr.* **2010**, *7*, 3.
[54] A. J. Bandodkar, P. Gutruf, J. Choi, K. Lee, Y. Sekine, J. T. Reeder, W. J. Jeang, A. J. Aranyosi, S. P. Lee, J. B. Model, R. Ghaffari, C. J. Su, J. P. Leshock, T. Ray, A. Verrillo, K. Thomas, V. Krishnamurthi, S. Han, J. Kim, S. Krishnan, T. Hang, J. A. Rogers, *Sci. Adv.* **2019**, *5*, eaav3294.
[55] A. J. Bandodkar, W. Z. Jia, C. Yardimci, X. Wang, J. Ramirez, J. Wang, *Anal. Chem.* **2015**, *87*, 394.
[56] J. Xiao, Y. Liu, L. Su, D. Zhao, L. Zhao, X. Zhang, *Anal. Chem.* **2019**, *91*, 14803.
[57] M. J. Buono, N. V. L. Lee, P. W. Miller, *J. Physiol. Sci.* **2010**, *60*, 103.
[58] B. Wang, X. M. Wen, P. Y. Chiou, N. T. Maidment, *Electroanalysis* **2019**, *31*, 1641.
[59] B. Wang, L. L. Feng, B. Koo, H. G. Monbouquette, *Electroanalysis* **2019**, *31*, 1249.


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